

Total state:  $\hat{\rho}_T$  on  $\mathcal{H}_S \otimes \mathcal{H}_E$

$$\hat{H}_{\text{int}} = \sum_i \hat{S}_i \otimes \hat{E}_i$$

System  $S$  ( $\hat{H}_S$ )

Environment  $E$  ( $\hat{H}_E$ )

$\text{Tr}_E$

Reduced state  $\hat{\rho}_S = \text{Tr}_E[\hat{\rho}_T]$